ConstructionSafety-FaithBench: Auditing Visual-Evidence Faithfulness in Construction-Safety Vision-Language Models

Abstract

Construction-safety vision-language models (VLMs) can produce the right safety answer for the wrong visual reason. A model may classify a scene as unsafe because construction context looks hazardous, while ignoring the specific worker, protective equipment, edge, or machine-proximity cue that the safety rule actually depends on. We introduce ConstructionSafety-FaithBench to audit this right-answer/wrong-evidence gap. The benchmark separates three questions that are usually merged: whether the answer is correct, whether the cited visual evidence is rule-relevant, and whether the answer changes when that evidence is removed. We evaluate automated baseline annotations and a Florence-2 grounding adapter on hard-hat, fall-protection, guardrail, and struck-by rules, then apply targeted occlusions to rule-relevant boxes and compare them with same-size random masks. On a prioritized audit set where Florence-2 often disagrees with the automated baseline, Florence reaches 18.3% accuracy and 12.5% macro-F1, while the baseline reaches 78.3% accuracy. More importantly, masking rule-relevant evidence changes Florence answers 39.2% of the time, compared with 9.2% for matched random masks. The result indicates that safety judgments can depend on specific local pixels, so answer scoring alone cannot establish visual-evidence faithfulness.